

I. Multiple-Choice Questions: Choose the most appropriate answer from the options provided. (25 POINTS)

2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25

1. Micheline currently manages a private cloud that has been built out in an on-premises data center. Some of the servers are reaching the end of their life and she has been tasked with finding a solution that minimizes the amount of capital investment necessary. Which of the following might she choose to implement for her company?
 - a. VPC
 - b. VNC
 - c. VPN
 - d. VDI
2. Morgan has hardened the virtual machines in the subnet he's just finished creating by turning off unnecessary services that were running in the background. However, he still wants to prevent unnecessary traffic from getting to the servers in the first place. Which of the following might he choose to implement in his cloud environment?
 - a. Virtual firewall
 - b. Internet gateway
 - c. VNC
 - d. Virtual switch
3. Erid is implementing new systems on a cloud service provider and needs to ensure that the finance systems and HR systems cannot communicate with each other without going through a virtual router. Which of the following should she implement?
 - a. Different cloud service providers for each department
 - b. An Internet gateway
 - c. Network address translation
 - d. Different subnets for each department
4. Angela has decided upon a multi-cloud deployment so that the company can continue to operate even if one of the cloud service providers experiences an outage. Which of the following explains the factor that was important to Angela in choosing this structure?
 - a. Cost optimization
 - b. Cloudbursting
 - c. Regulatory compliance
 - d. Disaster recovery
5. Cara works for a startup company that up until now has been running on a single virtual machine. The virtual machine is starting to see capacity issues and she is unable to provision more CPU and memory resources to satisfy its needs. Which of the following would allow her to expand the capacity of her company's web server?
 - a. DHCP
 - b. VPN
 - c. Load balancer
 - d. OSPF
6. Emina has three separate VPCs at the cloud service provider that her company uses. Some of the services in each of the VPCs need to communicate with some of the services in other VPCs on that same CSP. Which of the following might she implement in order to accommodate that requirement?
 - a. VLANs
 - b. Internet connection sharing
 - c. Cloud peering
 - d. IPsec
7. Fredrica is installing more infrastructure in her company's on-premises data center. She has installed 10 servers, but there is only one switch in the rack that the new equipment is going in. Which of the following technologies can she use to segment the servers into two groups without having to add a new switch that will also put them into separate broadcast domains?
 - a. VLAN
 - b. VPC
 - c. VNC
 - d. VLC
8. Lara is trying to decide on a protocol to use for a VPN to connect from one cloud service provider to another. She knows there are a variety of protocols out there that can be used for VPNs, but wants to make sure she chooses one that both cloud service providers support and that can be combined with IPsec. Which of the following protocols would she most likely need to choose?
 - a. L2TP
 - b. PPTP
 - c. GRE
 - d. PPP
9. Yolanda wants to see a list of MAC addresses that one of her servers has been communicating with. Which of the following command line tools will display that information?
 - a. telnet
 - b. arp
 - c. dig
 - d. ping

Tyrion wants to use a virtual firewall to filter the types of traffic that are allowed or not allowed into a virtual private cloud instance he has created on the cloud service provider his company uses. Which of the following can he use to accomplish this goal?

- a. VNC b. NAT c. NACL d. VLAN

Samwell wants to create a lightly protected subnet within the VNet that acts as a filter between the Internet and the other internal network resources. Which of the following describes the type of subnet that he is looking to deploy?

- a. ACL b. VPC c. NACL d. DMZ

Arya is looking to install a security appliance that is designed to detect applications and other resources running within the domain and monitor them according to her organization's policies. Which of the following would she want to implement?

- a. CASB b. CIFS c. CMDB d. CAB

A year ago Yvette configured two servers on separate VLANs. She still needs them to remain on separate VLANs, but now has a need for them to be able to send certain communications to each other over a certain port. Which of the following will be necessary for her to do?

- a. Migrate the servers to a VPC.
- b. Merge the VLANs together so that they both use two VLAN IDs.
- c. Create a route between the two VLANs so they know how to contact each other.
- d. Convert the 2 VLANs into a single VXLAN.

Jon recently returned from an IT conference where he learned about a technology that could alert the administrators to any intrusions that may occur by installing software on each of the servers within the cloud deployment. Unfortunately, this particular system wouldn't stop the intrusion automatically. Which of the following technologies did he learn about?

- a. HIPS b. HIDS c. NIPS d. NIDS

Joffrey is about to deploy a new web server. He wants to ensure that when a user accesses the server, that their web browsing session is encrypted between the browser and the server. Which of the following should he use with the web server to provide this functionality?

- a. SSH b. SSL c. TLS d. ACL

Theon wants to transfer some files to one of the Linux servers that he manages. Which of the following would ensure that the transmissions are secure while not relying on SSL/TLS for the encryption?

- a. FTP b. FTPS c. SFTP d. TFTP

Tormund has created a set of firewall rules and has noticed that when there is traffic that matches a rule, the traffic allowed in one direction automatically allows traffic in the other direction for an active connection as long as there is at least one message going in either direction within 10 minutes. What is the name of that type of firewall?

- a. Stateless b. Stateful c. Status-monitoring d. Active-analysis

Eddard wants to install a VM running in the perimeter network that provides antivirus/anti-malware capabilities for the rest of the network. Which of the following describes the type of VM that he wants to install?

- a. NSG b. NVA c. ASG d. NTP

Steven has been tasked with planning the move from an infrastructure where administrators and users add permissions to resources individually to one where users are assigned to groups based upon the department they work in and what they do in the department and then assigning the groups to the resources. Which of the following has he been tasked with planning?

- a. RBAC b. MAC c. JAC d. DAC

Fictional Corp has assigned each user within their Active Directory implementation a username. This username is a form of?

- a. Accounting b. Authentication c. Authorization d. Identity

Lincoln is assigning permissions to users and groups for a new application that he is deploying. As he's doing this, which of the following principles should he make sure he follows?

- a. Minimal overhead b. Least privilege c. Minimum access d. Least permission

22. Jeff has just been hired by Fictional Corp as a systems administrator. He is told that one of his responsibilities is to be the backup processes used for on-premises systems in the company's data center. He is told about documents that he should use to make sure that everything is done correctly. What type of documentation was he referred to?
- a. Baselines b. CMDB c. SOPs d. KPIs
23. Allen has been tasked with the design, development, and optimization of automation processes. Which of the following is the term that could be used to describe his current set of tasks?
- a. Configuration b. Management c. Orchestration d. Migration
24. Darius manages a directory of users for one of the domains within his organization. Which of the following describes the protocol in use for querying the directory?
- a. SDP b. LDAP c. AD d. NTLM
25. Otis wants to give a mobile app short-term access to a resource without having to store long-term user credentials (such as access keys) in the app. Which of the following describes the method he can use to do that?
- a. Permission b. Groups c. Policies d. Roles

Q2. Define the following: (4 POINTS)

A. Pre-coded PAMs:

B. Patchwork system:

C. Sticky sessions

D. Router-on-a-stick

13. Answer the following questions: (21 POINTS)

A. List three characteristics of NACL* (3 points)

B. List three characteristics of Security groups? (3 points)

C. List three limitations of VXLANs? (3 points)

D. Greta manages the on-premises networks for her company's Atlanta and Orlando offices. She hears about a technology that will allow her to connect certain network segments in Atlanta to their counterparts in Orlando by inserting the MAC address into layer 4 for UDP transport over the Internet. What is the technology that she has heard about that she should research more about in order to implement? (2 points)

Zaid Al-Laham

2025 first semester final chapters: chapter 1->7

Second semester: chapter 1->3(less than 10 marks in mcq only) and 4->6

7 was not included

1. Multiple-Choice Questions: Choose the most appropriate answer from the options provided. (25 POINTS)

1. Michonne currently manages a private cloud that has been built out in an on-premises data center. Some of the servers are reaching the end of their life and she has been tasked with finding a solution that minimizes the amount of capital investment necessary. Which of the following might she choose to implement for her company?

- a. VPC
- b. VNC
- c. VPN
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2. Morgan has hardened the virtual machines in the subnet he's just finished creating by turning off unnecessary services that were running in the background. However, he still wants to prevent unnecessary traffic from getting to the servers in the first place. Which of the following might he choose to implement in his cloud environment?

- a. Virtual firewall
- b. Internet gateway
- c. VNC
- d. Virtual switch

3. Enid is implementing new systems on a cloud service provider and needs to ensure that the financial systems and HR systems cannot communicate with each other without going through a virtual router. Which of the following should she implement?

- a. Different cloud service providers for each department
- b. An Internet gateway
- c. Network address translation

d. Different subnets for each department

4. Angela has decided upon a multi-cloud deployment so that the company can continue to operate even if one of the cloud service providers experiences an outage. Which of the following explains the factor that was important to Angela in choosing this structure?

- a. Cost optimization
- b. Cloud bursting
- c. Regulatory compliance

d. Disaster recovery

5. Cara works for a startup company that up until now has been running on a single virtual machine. The virtual machine is starting to see capacity issues and she is unable to provision more CPU and memory resources to satisfy its needs. Which of the following would allow her to expand the capacity of her company's web server?

- a. DHCP
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c. Load balancer

- d. OSPF

6. Ermina has three separate VPCs on the cloud service provider that her company uses. Some of the services in each of the VPCs need to communicate with some of the services in other VPCs on that same CSP. Which of the following might she implement in order to accommodate that requirement?

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7. Fredrica is installing more infrastructure in her company's on-premises data center. She has installed 10 servers, but there is only one switch in the rack that the new equipment is going in. Which of the following technologies can she use to segment the servers into two groups without having to add a new switch that will also put them into separate broadcast domains?

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8. Lara is trying to decide on a protocol to use for a VPN to connect from one cloud service provider to another. She knows there are a variety of protocols out there that can be used for VPNs but wants to make sure she chooses one that both cloud service providers support and that can be combined with IPsec. Which of the following protocols would she most likely need to choose?

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- c. GRE
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9. Yolanda wants to see a list of MAC addresses that one of her servers has been communicating with. Which of the following command line tools will display that information?

- a. nslookup
- b. arp
- c. dig
- d. ping

10. Tyron wants to use a virtual firewall to filter the types of traffic that are allowed or not allowed into a virtual private cloud instance he has created on the cloud service provider his company uses. Which of the following can he use to accomplish this goal?

- a. VNC
- b. NAT
- c. NACL
- d. VLAN

11. Samwell wants to create a lightly protected subnet within the VNet that acts as a filter between the Internet and the other internal network resources. Which of the following describes the type of subnet that he is looking to deploy?

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12. Arya is looking to install a security appliance that is designed to detect applications and other resources running within the domain and monitor them

according to her organization's policies. Which of the following would she want to implement?

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15. Jeffrey is about to deploy a new web server. He wants to ensure that when a user accesses the server, their web browsing session is encrypted between the browser and the server. Which of the following should he use with the web server to provide this functionality?

- a. SSH
- b. SSL
- c. TCP
- d. ACL

16. Thoen wants to transfer some files to one of the Linux servers that he manages. Which of the following would ensure that the transmissions are secure while not relying on SSL/TLS for the encryption?

- a. FTP
- b. FTPS
- c. SFTP**
- d. TFTP

17. Turmond has created a set of firewall rules and has noticed that when there is traffic that matches a rule, the traffic allowed in one direction automatically allows traffic in the other direction for an active connection as long as there is at least one message going in either direction within 10 minutes. What is the name of that type of firewall?

- a. Stateless
- b. Stateful**
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18. Eddard wants to install a VM running in the perimeter network that provides antivirus/antimalware capabilities for the rest of the network. Which of the following describes the type of VM that he wants to install?

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21. Lincoln is assigning permissions to users and groups for a new application that he is deploying. As he's doing this, which of the following principles should he make sure he follows?

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- a. Permission
- b. Policies
- c. Groups
- d. Roles

2. Define the following: (4 POINTS)

A. Pre-coded PAMs (chapter 7)

B. Patchwork system

Using parts of a security tool and parts of another security tool but this has a disadvantage as it leaves gaps in security coverage

C. Sticky sessions

A load balancing feature that ensures all requests from a single user are consistently sent to the same backend server for the duration of their session made to ensure no data loss

D. Router-on-a-stick

A network configuration where a single router is connected to a switch via one physical link to handle traffic between multiple different VLANs. It's a method for enabling communication across logically separated networks (inter-VLAN routing) without needing a separate physical router interface for each VLAN.

New question second semester final:

Colo (colocation) (instead of pre-recorded pams)

Used for creating a direct connection Connections offer low latency and assured bandwidth Connections do not rely on public Internet conditions

(Direct connection: Connection from the data center to the public cloud)

3. Answer the following questions: (21 POINTS)

A. List three characteristics of NACL? (3 points)

- NACL rules can either allow or deny traffic
- NACL rules are numbered for priority - start with lowest number
- Applicable rule is enacted immediately - no other rules viewed
- Rules are stateless - inbound and outbound traffic evaluated by separate rules

B. List three characteristics of Security groups? (3 points)

1. They only allow traffic: You can create rules to permit specific traffic, but you cannot create rules to explicitly deny traffic.
2. They are stateful: They automatically track connections, so if an outbound request is allowed, the corresponding return traffic is also allowed without needing a separate rule.
3. All rules are evaluated: When processing traffic, all rules are considered, and the most specific rule that matches the traffic type is the one that gets applied.

C. List three limitations of VXLANs? (3 points)

1. Increased Packet Overhead: VXLANs add extra headers to network packets, making them larger and increasing overall network traffic.
2. Requires Larger MTU: The larger packets require the physical network to be configured with a higher Maximum Transmission Unit (MTU) to prevent performance problems.
3. Greater Complexity: Its overlay/underlay architecture is more complex to design, configure, and troubleshoot than traditional VLANs.

For the second semester it was about VLANs instead of VXLANs

D. Greta manages the on-premises networks for her company's Atlanta and Orlando offices. She hears about a technology that will allow her to connect certain network segments in Atlanta to their counterparts in Orlando by inserting the MAC address into layer 4 for UDP transport over the WAN. What is the technology that she has heard about that she should research more about in order to implement? (2 points)

VXLAN (Virtual Extensible LAN)

E-> I weren't included in the second semester

E. Uma has been analyzing the performance and baselines from her virtual machines that are on a cloud service provider's platform. She sees that they do not appear to be performing up to the standards that her cloud service provider guarantees. What is a document outlines the standards that they are guaranteeing? (2 points)

SLA (Service Level Agreement)

F. Sarah has deployed a private cloud infrastructure that requires users to insert a smart card into their computer or into a card reader in order to authenticate them to use the applications. The smart card is associated with a certificate for each user, which is verified against a certificate authority. What has she deployed? (2 points) (chapter 7)

G. Alex is the network administrator for a Windows Server network. What is most likely in use to manage the users and other resources on that network? (2 points)
Active Directory (AD)

H. Noah has gone to work for the government and sees that some systems use a method of labeling data as classified, secret, and top secret rather than creating groups of users and applying permissions for those groups. What method is used in labeling data? (2 points)

MAC (Mandatory Access Control)

I. James works for a company where administrators and users add individual permissions to various resources. What access control methods is used by this company? (2. points)

DAC (Discretionary Access Control)

Questions that were also in the second semester final:

1.

Q2. Computational Questions (5 marks)

1. Adan is mapping out address pools for the subnets he plans to create in his Azure account. He has determined the network ID will use 192.169.100.0/24, and he needs 4 small subnets within this space. How many bits should Adan borrow from the host ID to create these subnets and what is the network ID of each subnet? (2 POINTS)

(2) $2^n \geq 4$
 $n = 2$ adan need to borrow two bits from host ID

Network ID = 192.169.100.0/26
① 192.169.100.0/26
② 192.169.100.64/26
③ 192.169.100.128/26

2. Douglas is setting up VM instances on a Hyper-V server to host some basic web server. He needs to know how many virtual machines could he create in each subnetwork?

Instead of /24 its /23.

2.

Gabriel has been given the 10.30.0.0 network and told to use the subnet mask 255.255.248.0 for the new cloud infrastructure he is building out for the subsidiary his company just purchased. Assuming

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C-Plus Chapter 4-6 | Quizlet

one of the IP addresses is used for a virtual router interface on that subnet, how many virtual machines could he create in each subnetwork?



Give this one a try later!

✕ Skipped

Correct answer

✓ c. 2045

3.

Negan has been given the 10.50.0.0/16 subnet to create the cloud infrastructure necessary for a new subsidiary that his company is creating, Zombies Inc. He wants to create a minimum of 500 smaller networks that can hold 100 servers each out of that space to minimize broadcast traffic within each subnet. Which of the following CIDR masks could he use to meet his requirements?

Give this one a try later!

✕ Skipped

Correct answer

quizlet.com/565257963/test?answerTermSides=6&promptTermSides=6&questionCount=96&questionTypes=1&showImages=true

2:50 PM

C-Plus Chapter 4-6 | Quizlet

✓ d./25

4. talked about an employee that had set up a web server connected to the cloud with public ip how to make users connect to it (check the answer)

1. using internet gateway to reach internet
2. making sure firewall isn't blocking necessary services like http (port 80)
Ssh (port22) https (port443)
3. making sure DNS is working